

FRC Framework v0.1

Baseline Reproduction and Regression

Weekly Research Update

Juan Pablo Solís Ruiz

USTC — Supervisor: Prof. Ren Haijun

September 29, 2026

Purpose of This Update

Goal

Report the completion of the first reproducible FRC simulation workflow: structured configuration, input generation, SCNet execution, and regression against the inherited WarpX baseline.

- The inherited 2000-step formation case was reproduced through Framework v0.1.
- The new run was compared against two frozen legacy realizations.
- Both scalar and spatial diagnostics show agreement with the inherited baseline.
- Next decision: whether to extend the formation case to 4 μs , after preserving the validated baseline.

Framework v0.1: What Was Built

Validated workflow

TOML → validation → WarpX input → manifest
→ Slurm → diagnostics

- Structured case configuration.
- Human-readable generated WarpX input.
- SCNet Slurm script generation.
- Hash-based provenance manifest.
- Pre-submission integrity checks.

```
frc_framework_v0_1/  
  configs/formation/  
    baseline.toml  
    smoke.toml  
  templates/  
    warpx_formation.in  
    scnet_cpu.sbatch  
  src/frc/  
    config.py  
    validation.py  
    input_generator.py  
    manifest.py  
  scripts/  
    generate_case.py  
    submit_case.py  
  runs/
```

Workflow Validation: Smoke Test

Smoke case

$$N_t = 20, \quad N_{\text{ppc}} = 20, \quad t_{\text{end}} = 0.02 \mu\text{s}$$

- Same geometry, fields, timestep, solver, and boundaries as the inherited case.
- Checkpoints disabled for the smoke test.
- Purpose: validate the software and HPC pipeline, not FRC formation physics.

Result

- Slurm job completed.
- Exit code: 0:0.
- 4 MPI ranks $\times$ 8 OpenMP threads.
- Diagnostics produced: diag000000, diag000020.
- No checkpoint directories.

Framework v0.1 smoke test: PASS

Full Baseline Run Through the Framework

Full regression case

$$N_t = 2000, \quad N_{\text{ppc}} = 200, \quad t_{\text{end}} = 2 \mu\text{s}$$

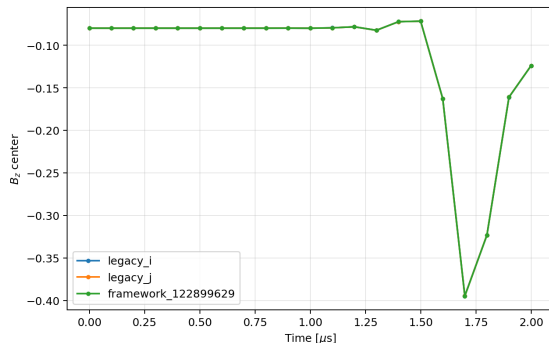
- Slurm job: 122899629.
- Node: i08r4n07.
- Runtime: 02:36:33.
- Exit code: 0:0.
- Output size: approximately 12 GB.

Output structure

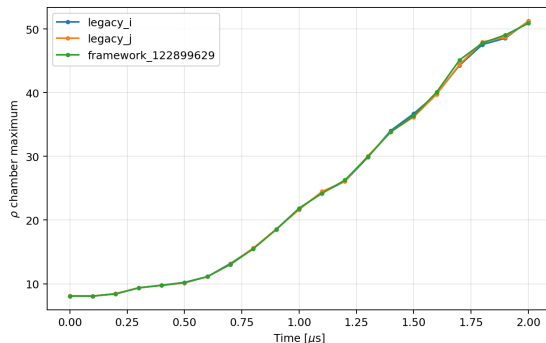
- 21 plotfiles:
 $\text{diag000000} \rightarrow \text{diag002000}$
- 2 checkpoints:
 $\text{chk000000}, \quad \text{chk002000}$
- warpx_used_inputs
- runtime provenance record

The wall-clock time and output structure match the inherited workload.

Scalar Regression: Magnetic Field and Density



Central B_z

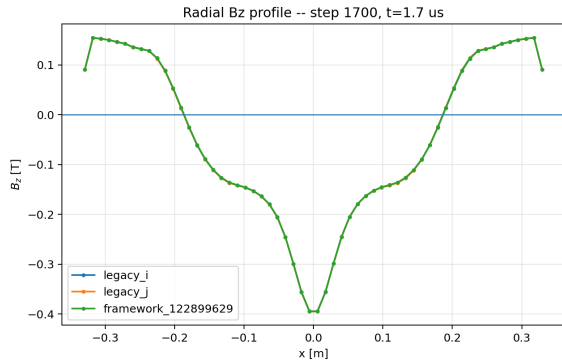


Maximum chamber density diagnostic

Maximum relative deviations from legacy midpoint

$B_{z,\text{center}}$: 0.082%, $B_{z,\text{min}}$: 0.71%, $B_{z,\text{max}}$: 0.23%, ρ_{max} : 1.73%.

Spatial Regression Near Maximum Compression



Magnetic-field agreement

At $t = 1.7 \mu$ s:

$$\frac{\|B_z^{\text{fw}} - B_z^{\text{legacy}}\|_2}{\|B_z^{\text{legacy}}\|_2} = 2.71 \times 10^{-3}$$

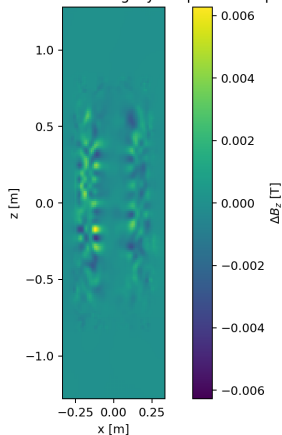
$$\text{corr}(B_z) = 0.999996$$

$$\text{sign agreement} = 99.991\%.$$

- Same field-reversal proxy as legacy runs.
- Same radial structure at the strongest compression stage.

Residual Field at the End of the Baseline

Bz difference: framework - legacy midpoint -- step 2001



$$B_z^{\text{framework}} - \frac{1}{2}(B_z^i + B_z^j) \text{ at } t = 2.0 \mu\text{s}$$

End-state spatial comparison

$$B_z \text{ } L_2 \text{ difference} = 5.19 \times 10^{-3}$$

$$\text{corr}(B_z) = 0.999986$$

$$\rho \text{ } L_2 \text{ difference} = 3.68\%$$

$$\text{corr}(\rho) = 0.99907.$$

- Magnetic geometry remains essentially identical.
- Density differences are small and spatially correlated.
- T_i is noisier and remains an informative-only metric.

Regression Conclusion

Result

Framework v0.1 inherited-baseline regression: PASS

Evidence:

- 111 active WarpX input assignments reproduced.
- Same mesh, timestep, solver, field drive, diagnostics, and checkpoints.
- Successful SCNet execution with the inherited MPI/OpenMP layout.
- Scalar magnetic diagnostics agree at sub-percent level.
- Spatial B_z agrees within $< 0.52\%$ relative L_2 .
- Field-reversal proxy evolves identically: absent at $0 \mu\text{s}$, present afterward.

Interpretation: the framework has not introduced a detectable systematic change to the inherited macroscopic formation behavior.

Next Step: Extend or Move to GS/MRR?

Option A: Extend formation to 4 μs

Motivation:

$$t_{\text{peak}} = 4 \mu\text{s}, \quad t_{\text{baseline}} = 2 \mu\text{s}.$$

The inherited case ends before the imposed magnetic-field ramp reaches its nominal peak.

- Useful for understanding full formation.
- Needs storage/runtime planning.
- Should use the validated framework, not manual edits.

Option B: Move to GS/MRR + tilt

Immediate scientific questions:

- Alfvén-time normalization.
- S^* definition.
- deuterium vs proton mass in d_i .
- $n = 1$ tilt growth rate.
- velocity/current diagnostic interpretation.

Recommendation: preserve the validated formation baseline and discuss whether the next priority is a 4 μs formation run or the GS/MRR tilt case.

Summary

- Framework v0.1 now generates, validates, submits, and tracks FRC WarpX runs.
- A 20-step smoke test passed end-to-end.
- The full inherited 2000-step case completed successfully through the framework.
- Scalar and spatial regression tests reproduce the legacy macroscopic behavior.
- The inherited formation baseline is now a validated regression reference for future changes.

Next decision: $4\ \mu\text{s}$ formation extension or GS/MRR tilt reproduction.